

VIRTUAL PATCHING 5.2.0

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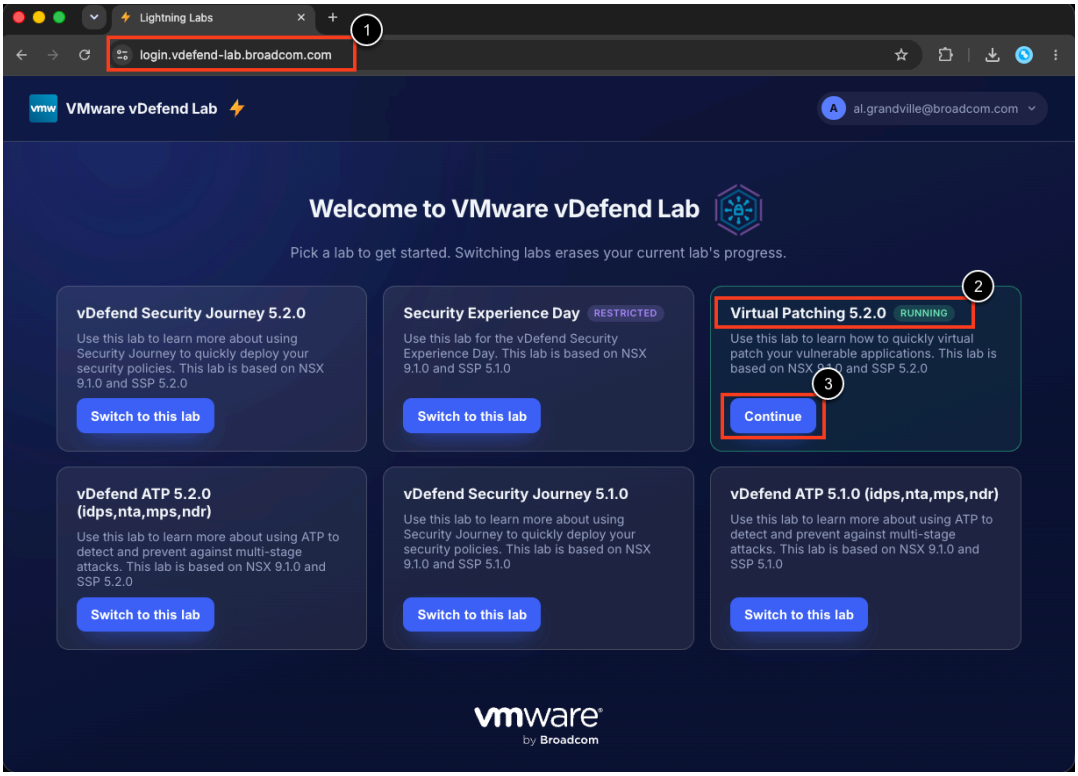
The urgency of implementing compensating mechanisms such as Distributed Virtual Patching is amplified by modern AI models rapidly discovering zero-day exploits. To maintain optimal throughput, enforce these Intrusion Detection and Prevention System (IDPS) signatures directly within the emergency policy engine at the top of the evaluation stack. Placing targeted dynamic rules at the top of the inspection baseline prevents redundant evaluation cycles across broader threat profiles. Dynamic virtual patches must be engineered precisely to block malicious payload vectors while eliminating false positives on legitimate application traffic. Achieving this level of precision requires rigorous signature selection to safeguard production workloads from accidental service interruption. Security teams should restrict rule scopes by defining explicit Source and Destination criteria whenever possible. Furthermore, leverage granular Service parameters to pinpoint the exact network ports and protocol vectors leveraged by the attack payload. Enforcing these inspection policies directly at the hypervisor-embedded virtual network interface card (vNIC) isolates attack vectors before exploit traffic reaches host workloads or compromises unpatched operating system kernels. This distributed edge enforcement shields high-risk vulnerabilities, granting engineering teams the operational runway necessary to code, test, and ship permanent software updates without incurring emergency application downtime.

Lab Objective: Demonstrate the discovery of a zero-day/unpatched vulnerability within the DataCenter. Complete a successful exploitation of the vulnerability and show that the IDPS control (in detect only mode) was in fact able to detect this exploit. Once the vulnerabilities are known and understood, construct a Distributed Virtual Patching ruleset to detect and prevent any future exploits from being successful and giving the application team ample time to identify, download, implement and test the patch.

Lab Topology & Component Matrix

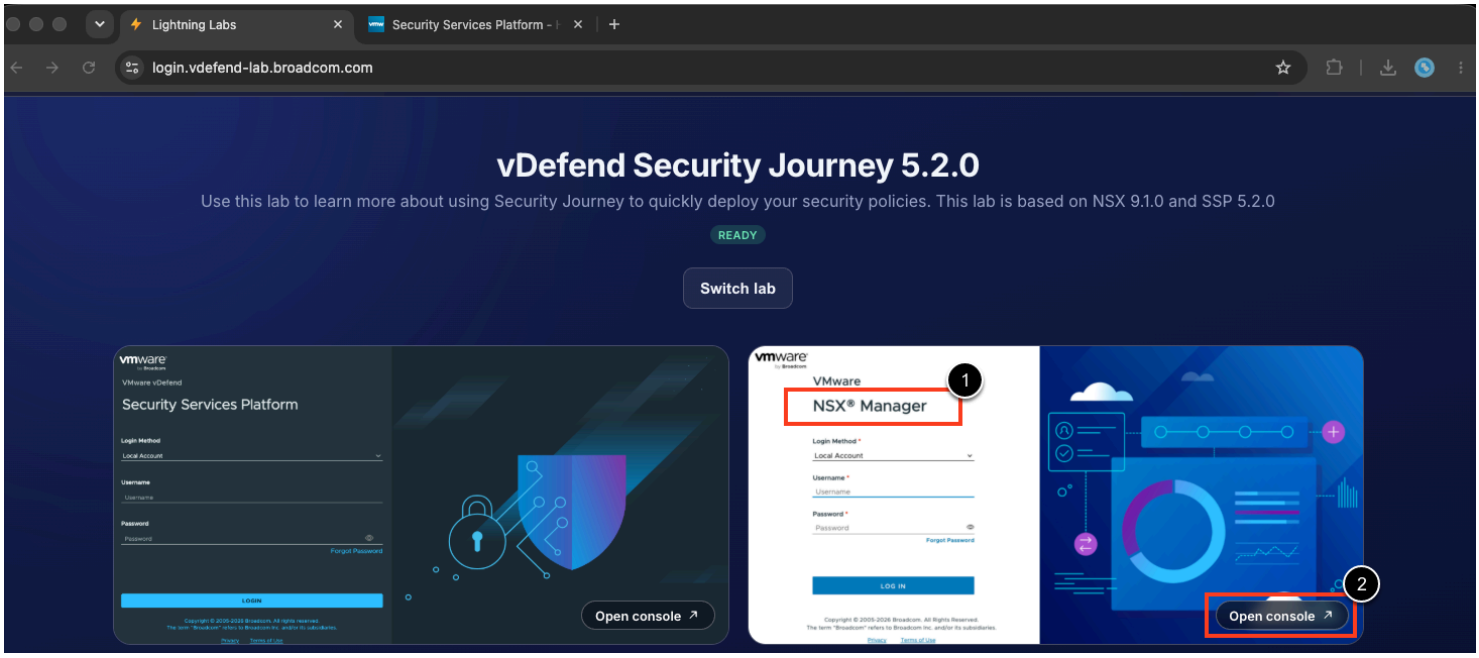
Component Role	Specification / Image	Description & Function
Attacker Host	Linux / Kali	Simulates targeted exploit vectors and payloads against vulnerable application endpoints.
Compensating Control	vDefend Distributed IDPS	Security control that sits inline with the Virtual Network Interface of the workloads regardless of the network topology
Target Workload	n8n is a node-based workflow automation platform that lets you connect different apps, APIs, and AI models to automate repetitive tasks	An Unpatched version will discovered using a vulnerability scanner within the datacenter
Vulnerability Scanner	Nessus will be used to test for and discover the n8n virtual machine in the datacenter	Nessus is one of the most used vulnerability scanners in the industry to search for vulnerabilities in the environment.

Welcome to the VMware vDefend Lab



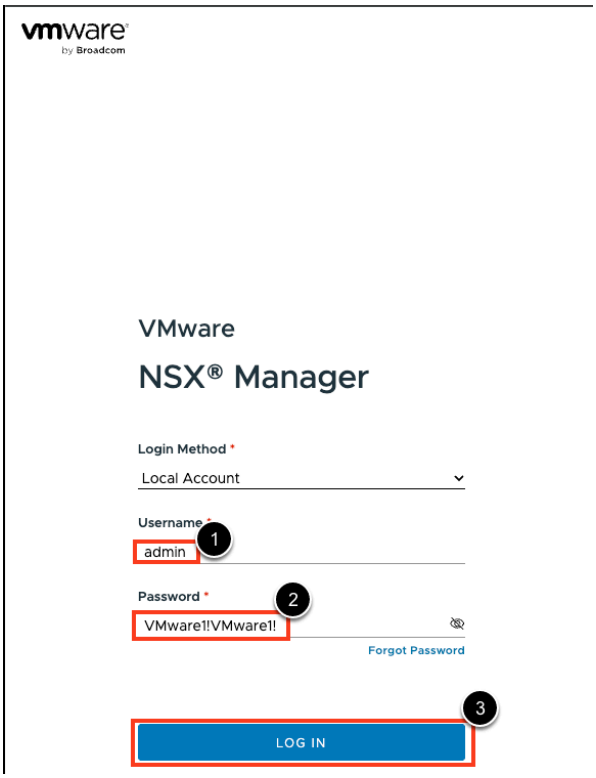
1. Lab URL = https://login.vdefend-lab.broadcom.com/
2. Locate the Virtual Patching **5.2.0** tile
3. Click "**Switch to this lab**" or "**Continue**"

Launch the NSX Manager Console



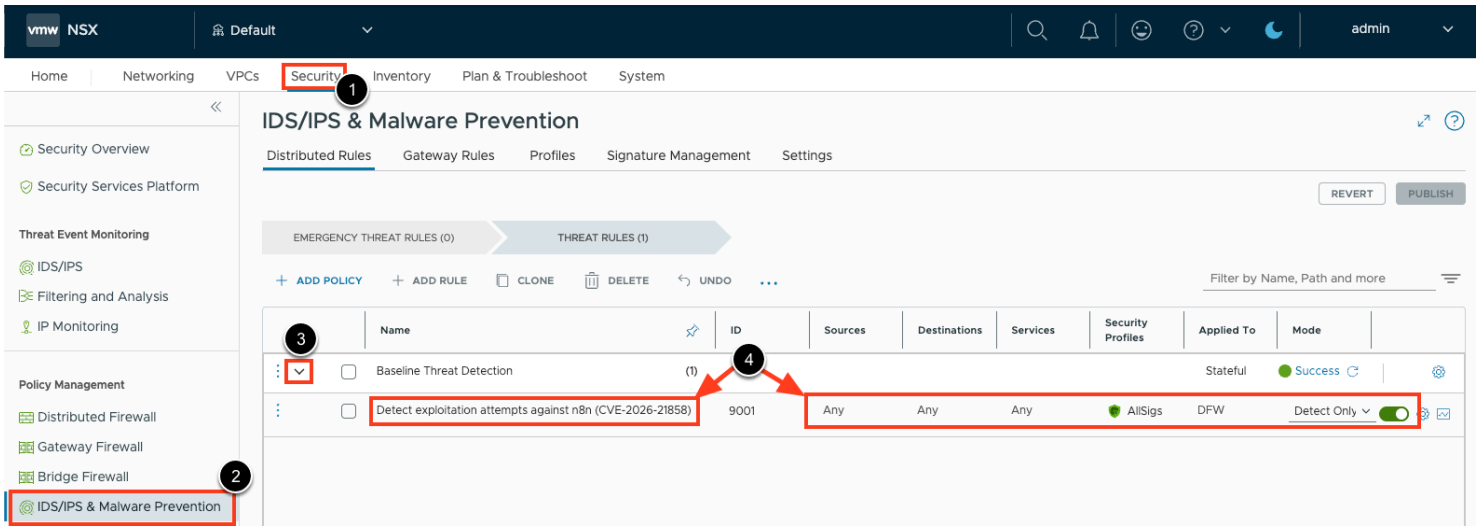
1. Locate the **NSX Manager** tile
2. Click **Open console**

Login to the NSX Manager



1. Username = **admin**
2. Password = **VMware1!VMware1!**
3. Click **LOGIN**

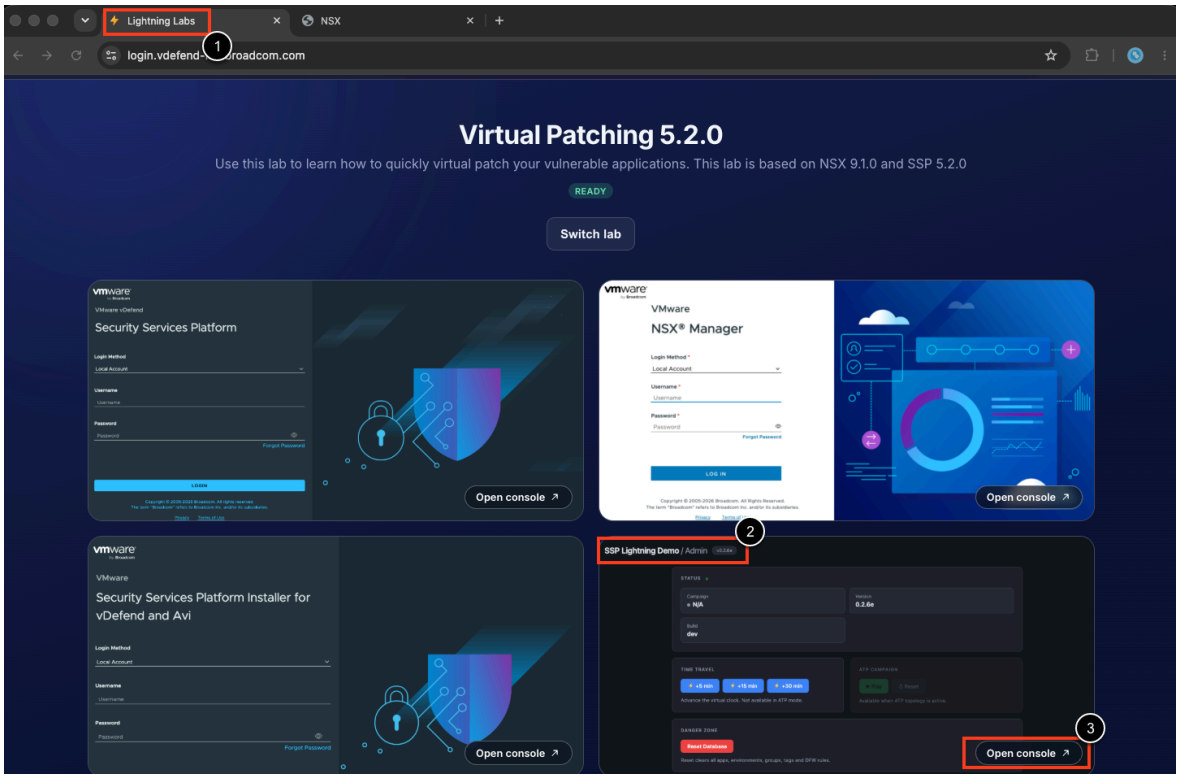
Examine the Example IDPS Policy



1. Click **Security**
2. Click **IDS/IPS & Malware Prevention**
3. Click the **arrow** to the left of **Baseline Threat Detection** policy for additional detail
4. Review the **Detect exploitation attempts against n8n (CVE-2026-21858)** rule
    - Sources = **Any**
    - Destinations = **Any**
    - Services = **Any**
    - Security Profile = **AllSigs**
    - Apply To = **DFW**
    - Mode = **Detect Only**

> This policy will detect all known threats against the workloads.
>
> This is a very coarse grained policy but will effectively detect known threats against the workloads.
>
> This policy is applied to every workload in the environment.

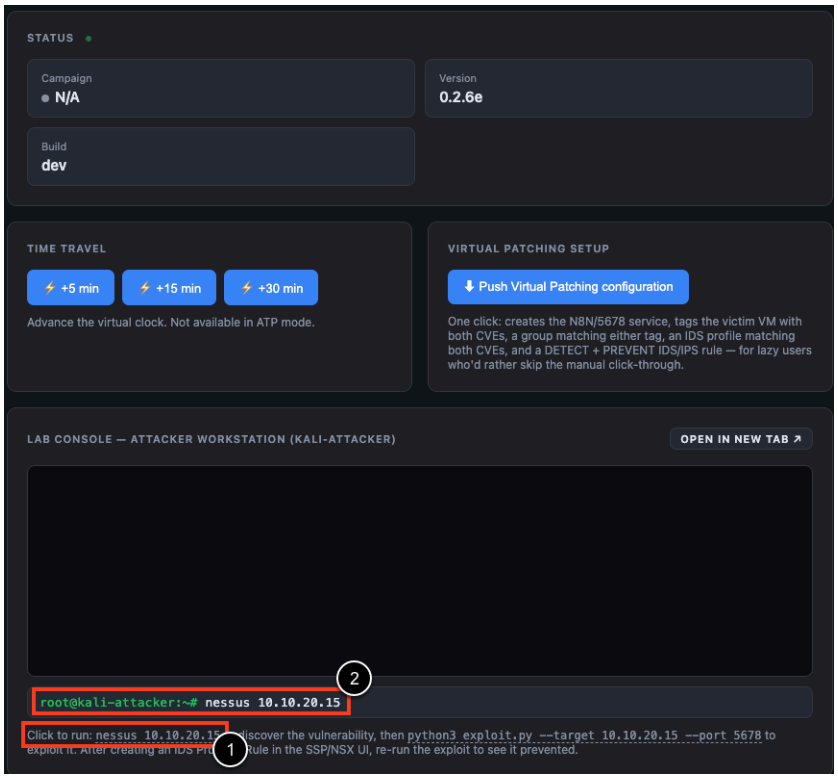
Setup the Attacker Environment



1. Switch to the **Lightning Labs tab**

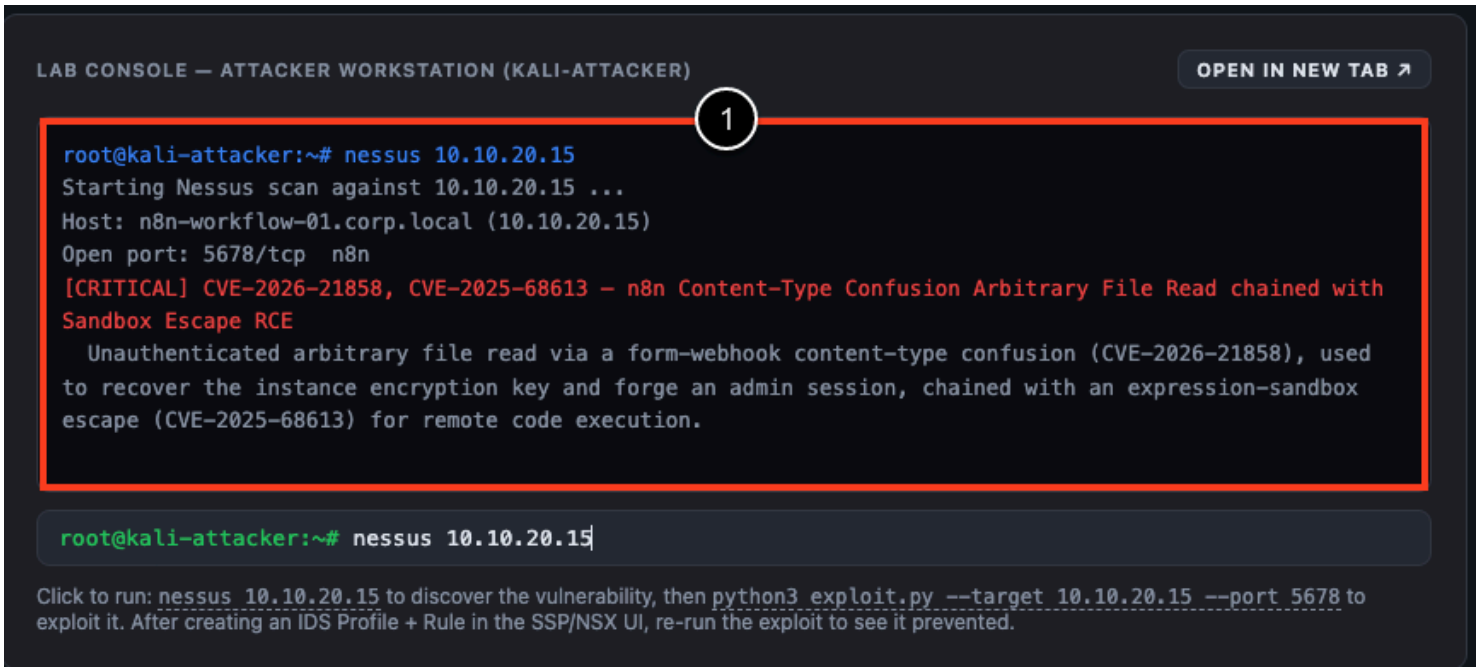
2. Locate the **SSP Lightning Demo / Admin tile**
3. Click **Open console**

Run a Nessus scan on the n8n server



1. **Click to run: nessus 10.10.20.15** (this will auto type the command into the console)
2. Review the **command** in the console and press **enter**

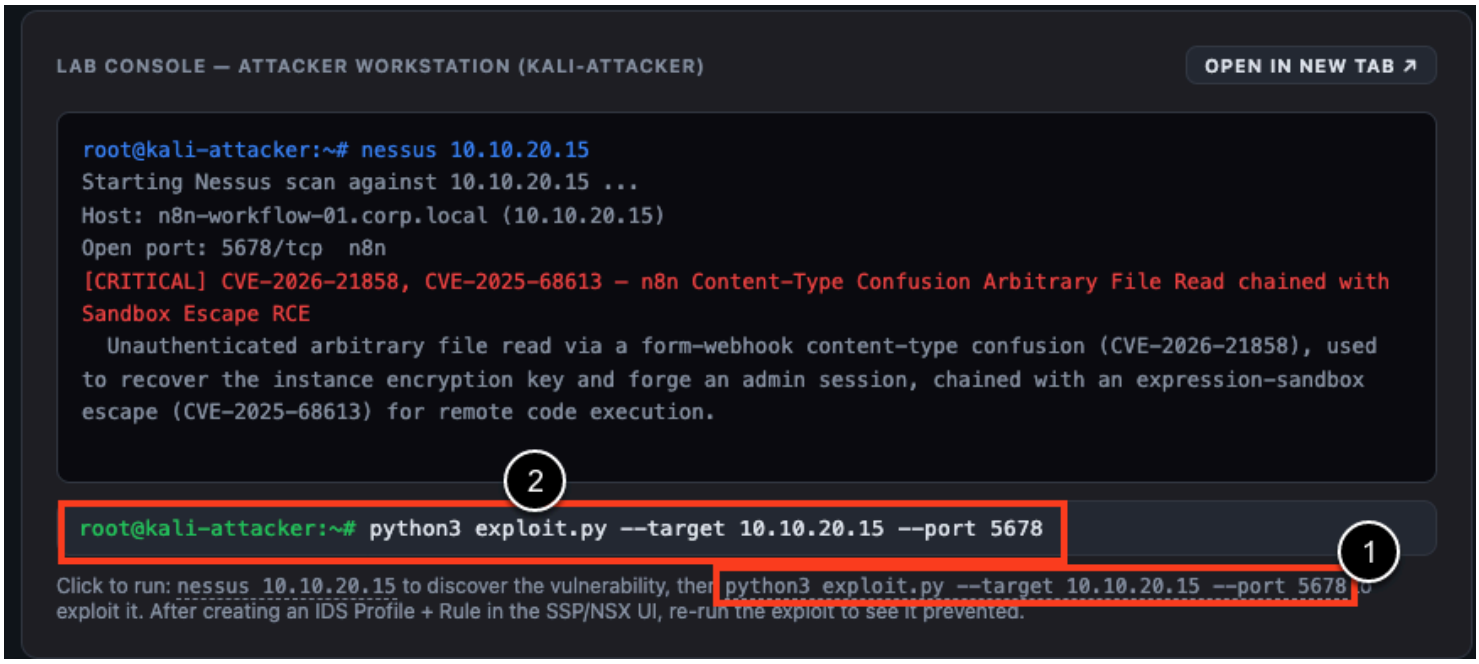
Review Nessus scan results



1. Review the **results** of the Nessus scan

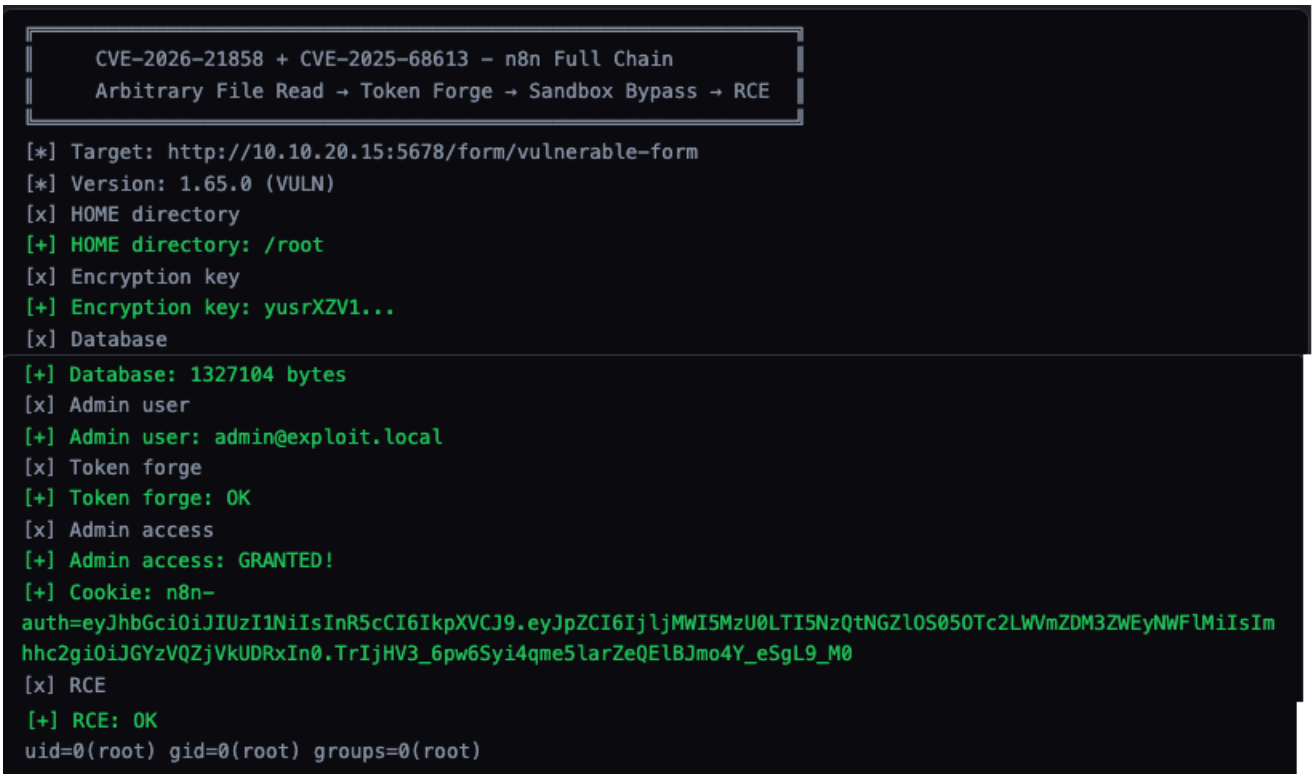
> You as well as the attacker will quickly discover that the n8n server is in fact vulnerable to these vulnerabilities.
>
> Click here for more information on the Vulnerabilities: CVE-2025-68613 CVE-2026-21858
>
> Consider searching the KEV list to confirm that these exploits are known to be in use at this time.

Exploit the n8n Server



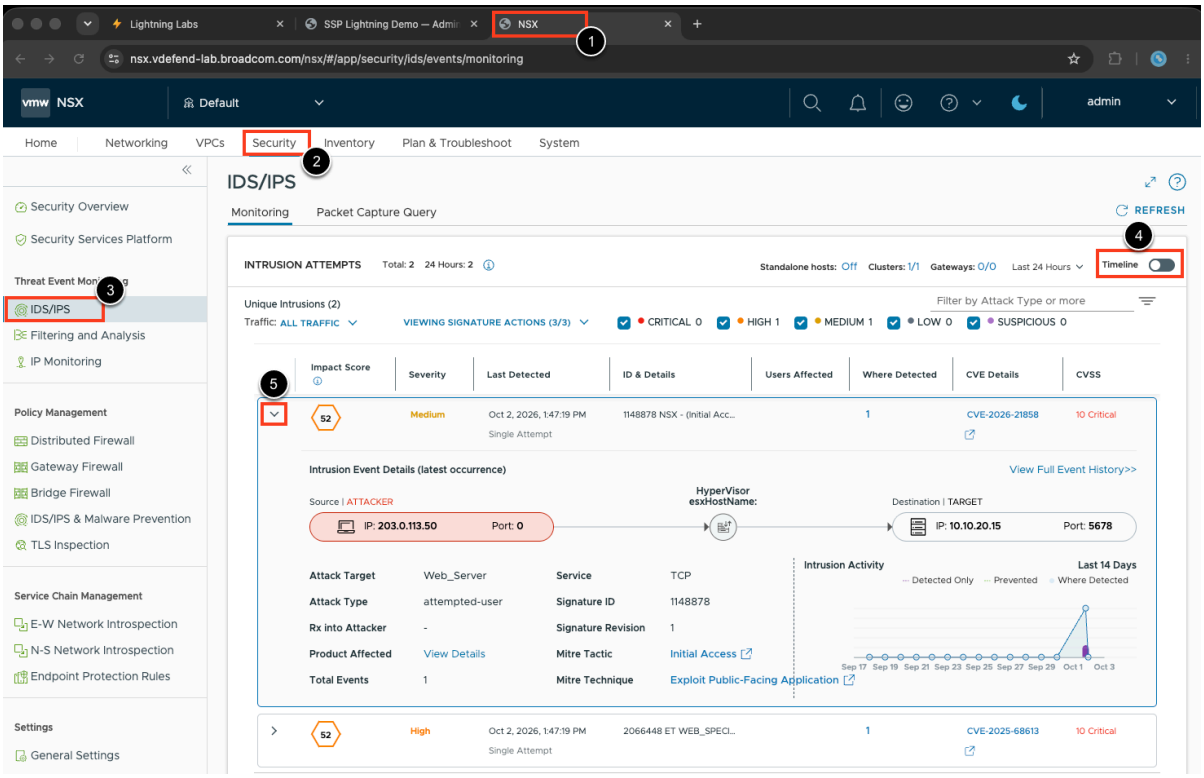
1. **Click to run: python3 exploit.py --target 10.10.20.15 --port 5678** (this will auto type the command into the console)
2. Review the **command** in the console and press **enter**

Review the Exploit



You were in fact able to successfully exploit the n8n server for full (root) control. Please **do not** do this anywhere else as it is always important to have permission to use tools like these!!!

Examine the IDS Alerts



1. Switch to **NSX Manager**
2. Click **Security**
3. Click **IDS/IPS**
4. **Hide** the **timeline** to create more screen space to review the events
5. Click the **arrow** next to the event to view details

You have successfully found and exploited a vulnerability within the lab setup!

Determine the appropriate coverage with IDPS Signatures

As Virtual Patches represent a compensating control that will be installed in a Prevention Mode, it is recommended to be as specific as possible in the IDPS Ruleset.

Pro Tip: You can also use **vTIS** to do research on the available protections (IDS Signatures) within the vDefend Product.

Consider the existing **Detect Only** Ruleset

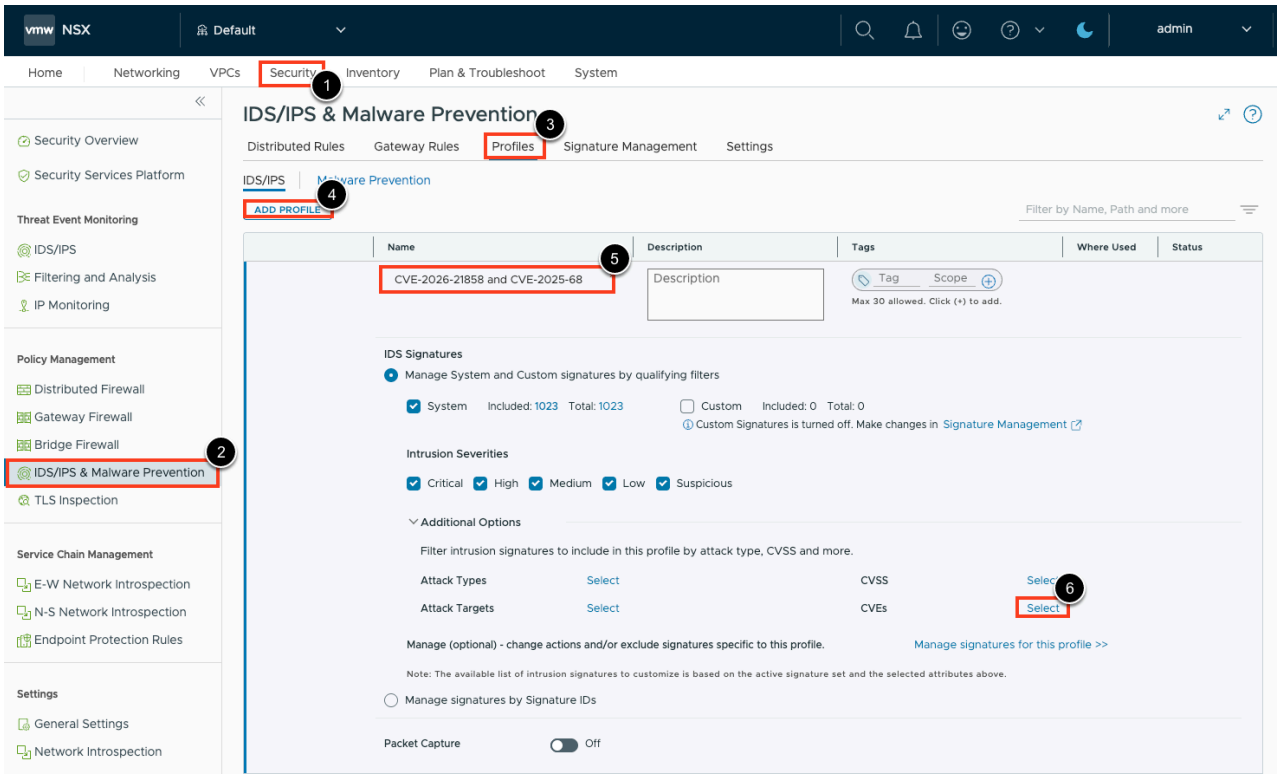
- Sources = **Any**
- Destinations = **Any**
- Services = **Any**
- Security Profile = **AllSigs**
- Apply To = **DFW**

We have learned a great deal about the attack and have all of the useful details to create a much more specific ruleset.

A more specific rule could contain:

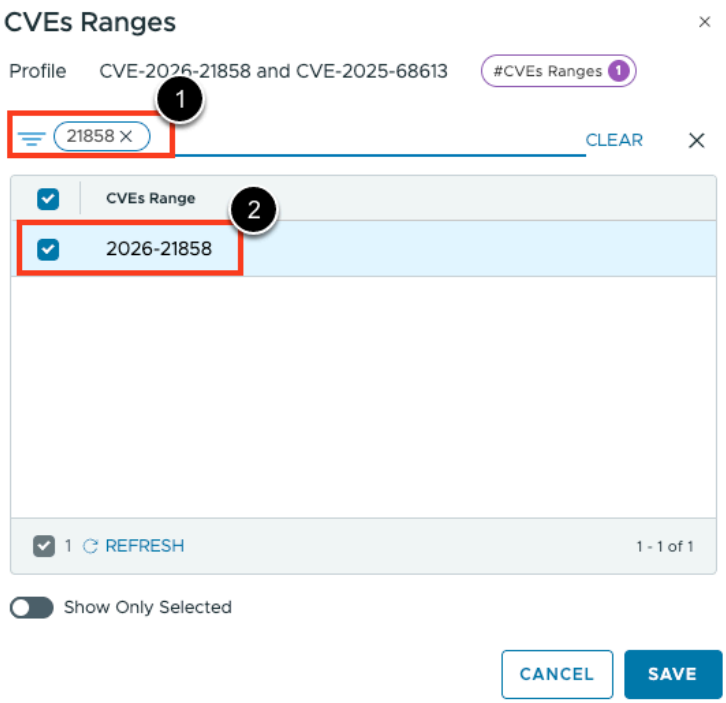
- Sources = **Any**
- Destinations = **n8nVM**
- Services = **n8n (tcp 5678)**
- Security Profile = **Virtual Patch: CVE-2026-21858 and CVE-2025-68613**
- Apply To = **n8nVM**

Add a new IDS/IPS Profile



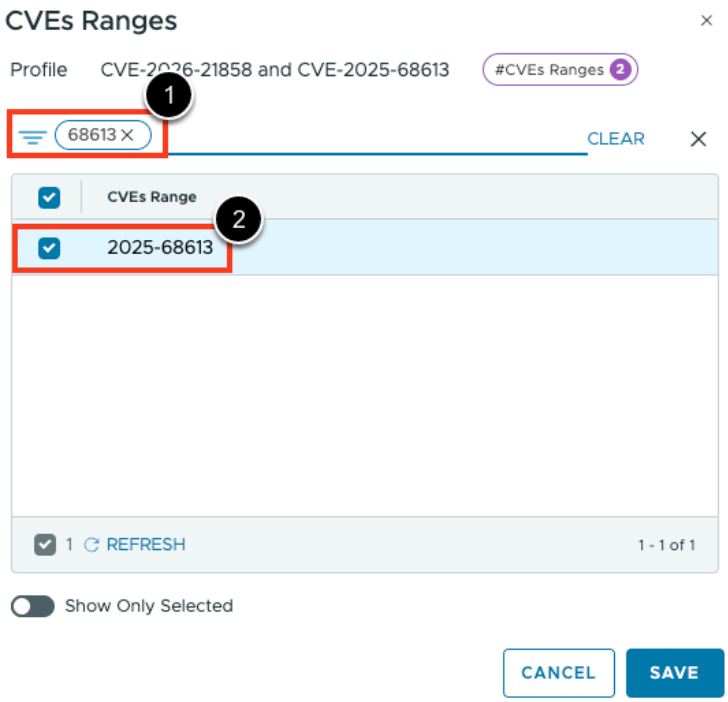
1. Click **Security**
2. Click **IDS/IPS & Malware Prevention**
3. Click **Profiles**
4. Click **ADD PROFILE**
5. Click in the Name field and type **CVE-2026-21858 and CVE-2025-68613**
6. To the right of CVEs, click **Select**

CVEs Ranges



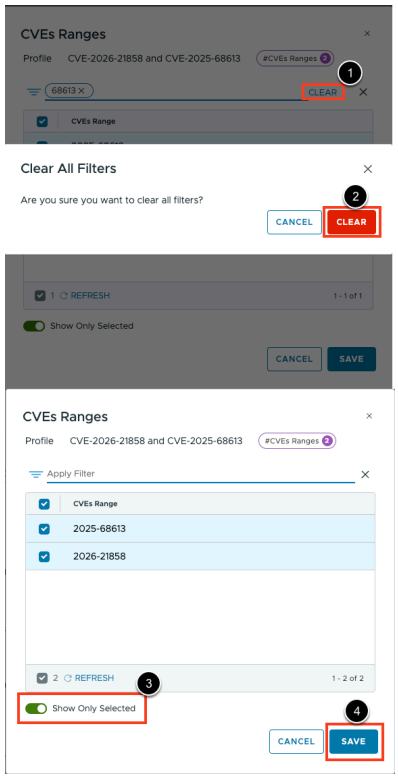
1. Click in the **Apply Filter** box and type **21858** and press **enter**
2. Select **2026-21858**

CVEs Ranges



1. Click in the **Apply Filter** box and type **68613** and press **enter**
2. Select **2025-68613**

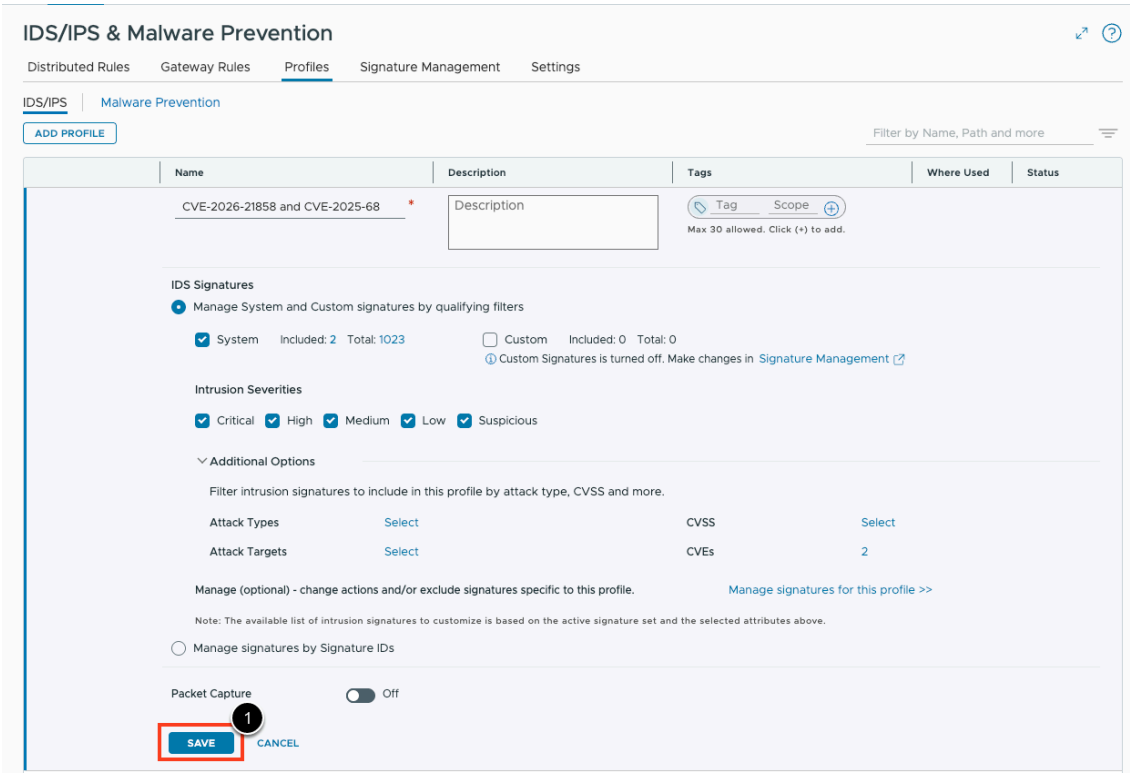
CVEs Ranges



1. Click **CLEAR**
2. Clear All Filters -> Click **CLEAR**
3. Enable **Show Only Selected**
4. Click **SAVE**

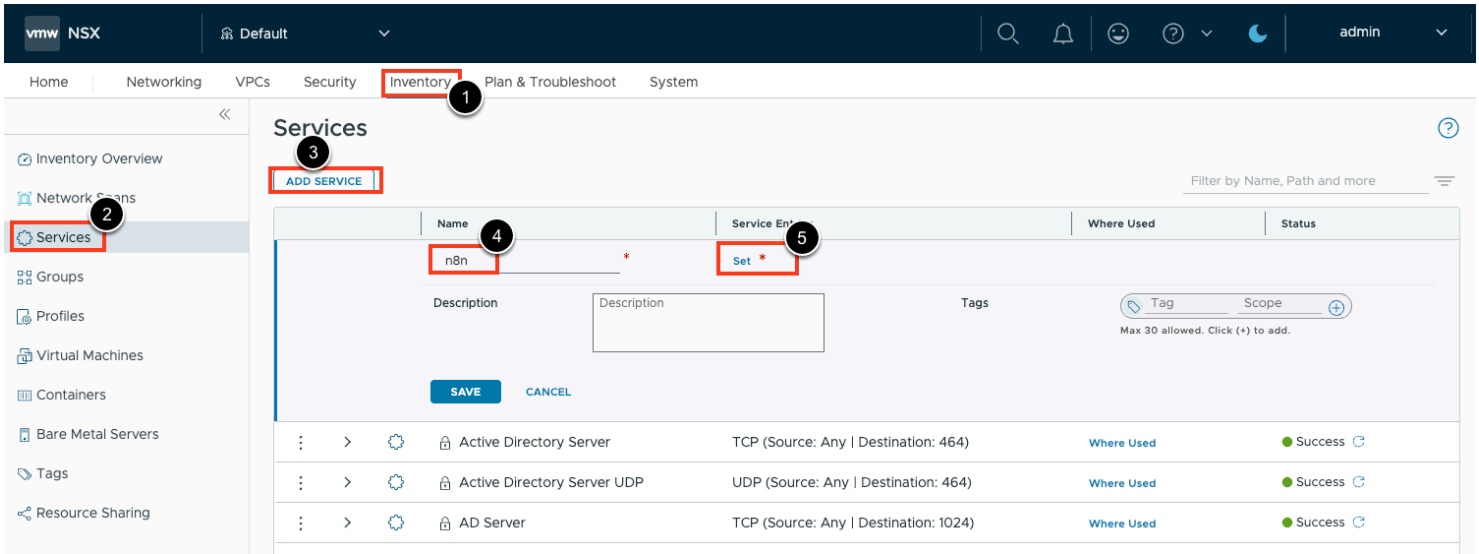
Starting in the 9.1.0 release, the option to search by CVE was added. This will allow you to search for and select the signatures that protect against these specific attacks.

Save IDS/IPS Profile



1. Click **SAVE**

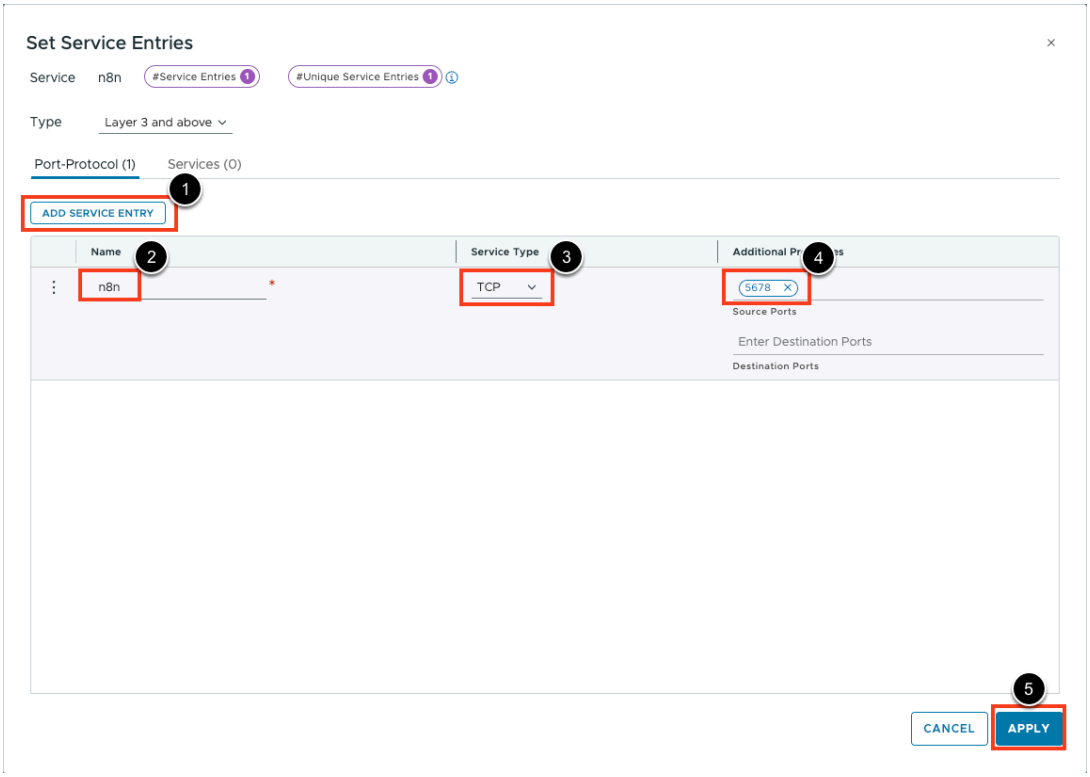
Add a Service object for n8n



1. Click **Inventory**

2. Click **Services**
3. Click **ADD SERVICE**
4. In the name field, type **n8n**
5. Click **Set**

Set Service Entries



Click **ADD SERVICE ENTRY**

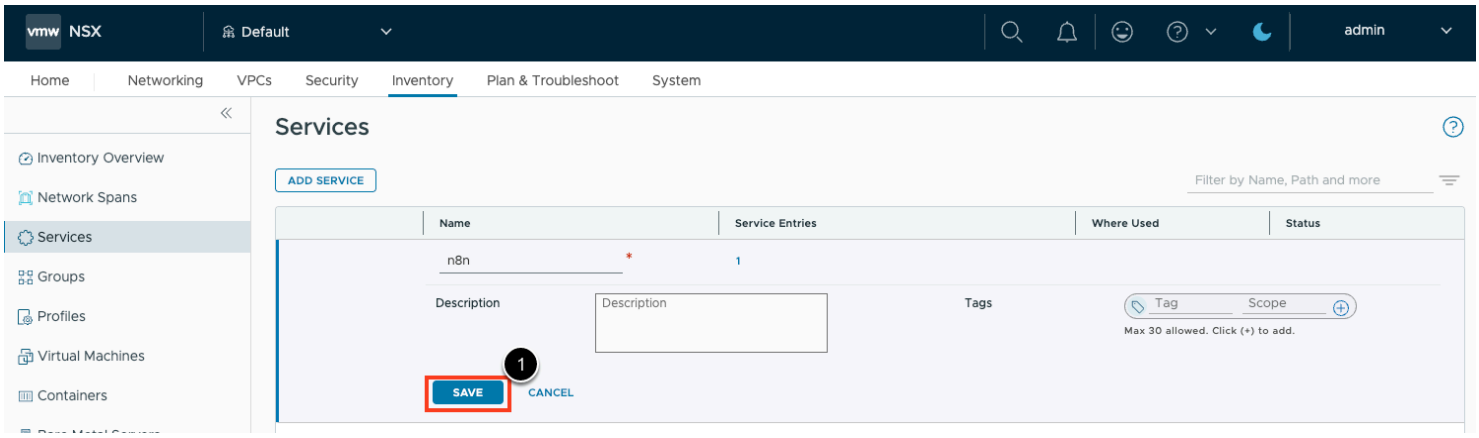
In the name field, type **n8n**

Service Type = **TCP**

Source Ports = **5678**

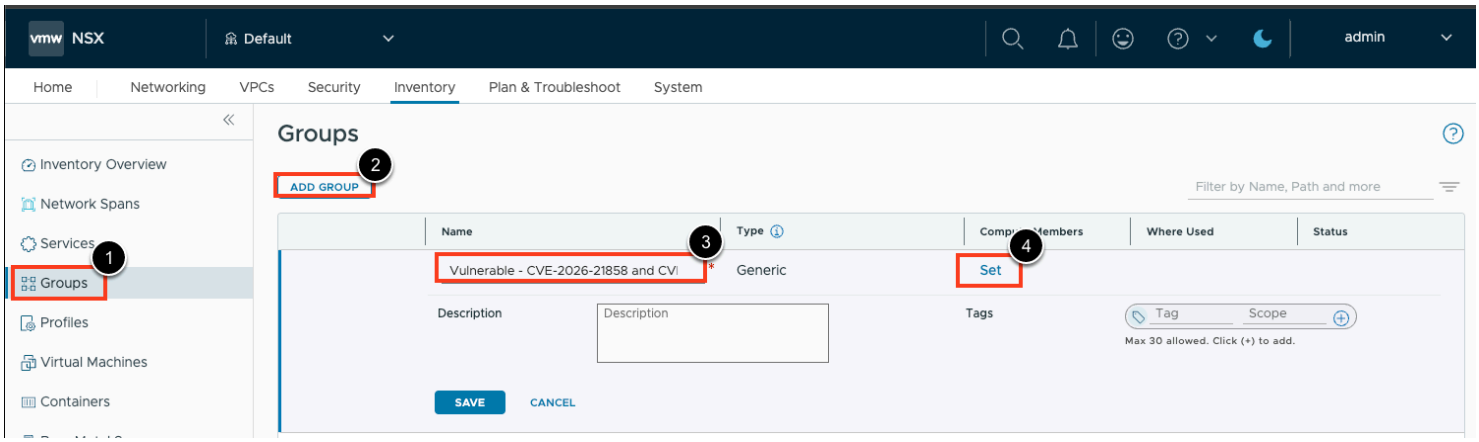
Click **APPLY**

Save the n8n Service object



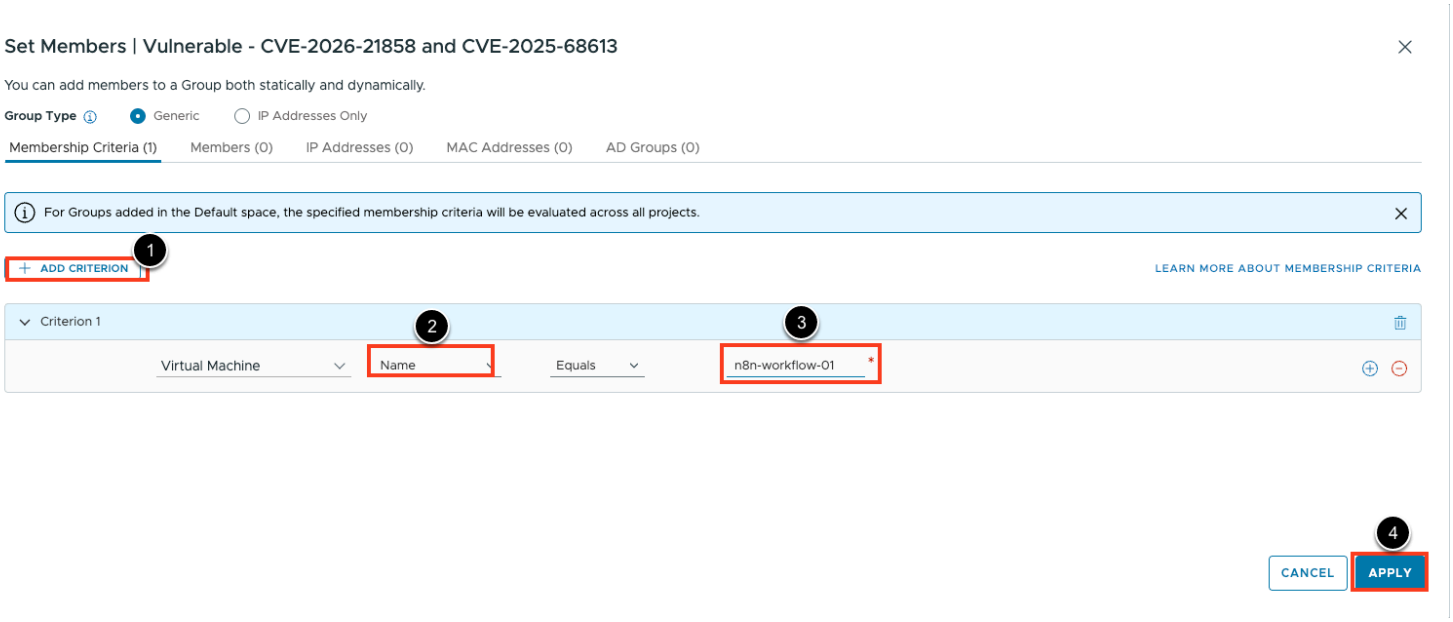
1. Click **SAVE**

Create an NSX Group for the n8n VMs



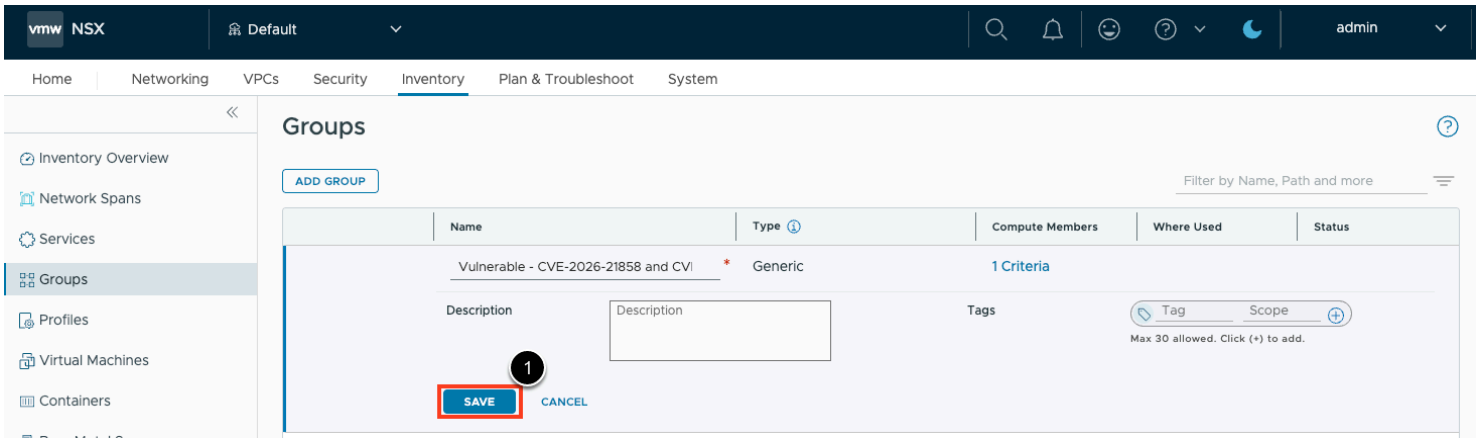
1. Click **Groups**
2. Click **ADD GROUP**
3. In the name field, type **Vulnerable - CVE-2026-21858 and CVE-2025-68613**
4. Click **Set**

Set Members | Vulnerable - CVE-2026-21858 and CVE-2025-68613



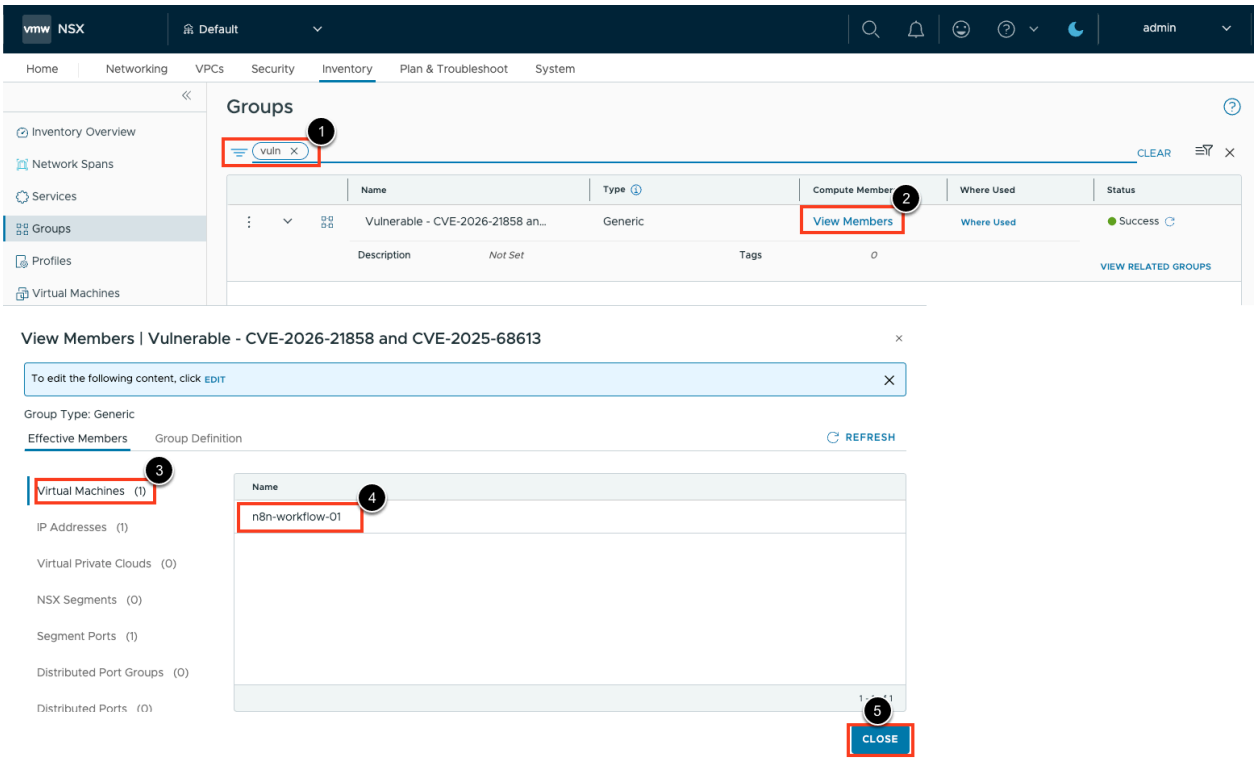
1. Click **ADD CRITERION**
2. Select **Name**
3. Type in **n8n-workflow-01**
4. Click **APPLY**

Save Group



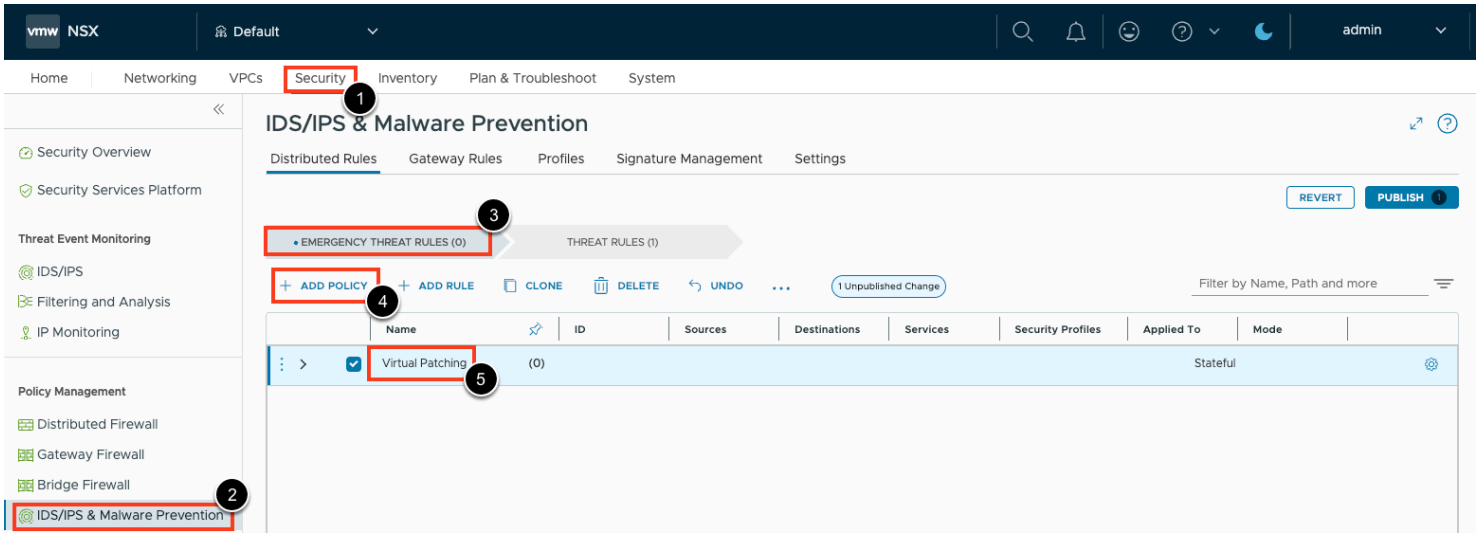
1. Click **SAVE**

Verify Group Members



1. In the **Apply Filter** field, type **vuln** and press **enter.**
2. Click **View Members**
3. Click **Virtual Machines (1)**
4. Note that the VM named **n8n-workflow-01** was added to the group dynamically.
5. Click **CLOSE**

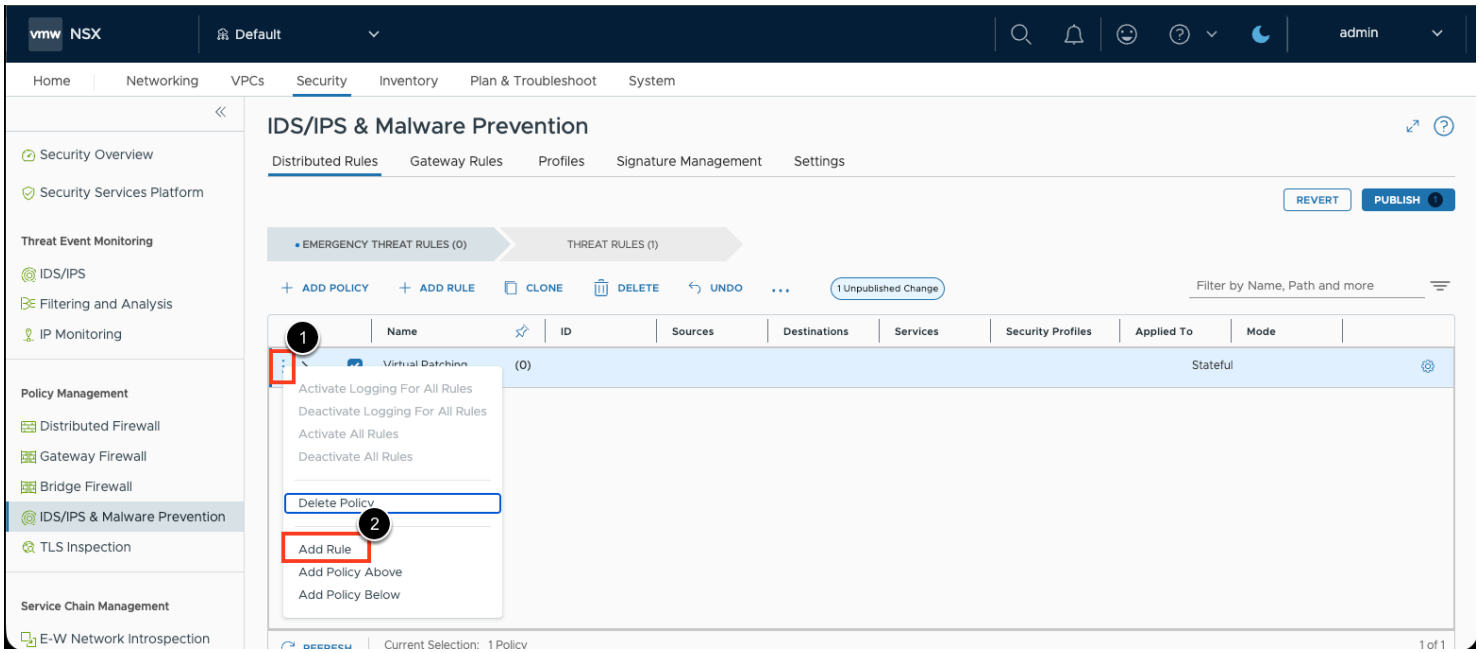
Create Virtual Patching Policy



1. Click **Security**
2. Click **IDS/IPS & Malware Prevention**
3. Click **EMERGENCY THREAT RULES**
4. Click **ADD POLICY**
5. In the name field, type **Virtual Patching** and press **enter**

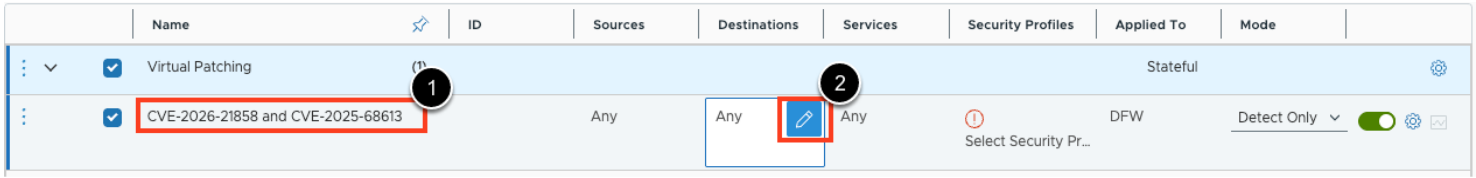
** Note ** It is recommended to place these rules in the Emergency Section of IDPS RuleTable. These very specific rules can be easily shadowed by more broad rules like the one we looked at prior. Placing these in the Emergency Section ensures that these specific rules are considered first and will not prevent other rules from being considered in the Threat Rules Section. These rules are evaluated in the exact same way that a Firewall rule would be considered. (Left to Right and Top Down in each section) stopping on the first match.

Create Virtual Patching Rule



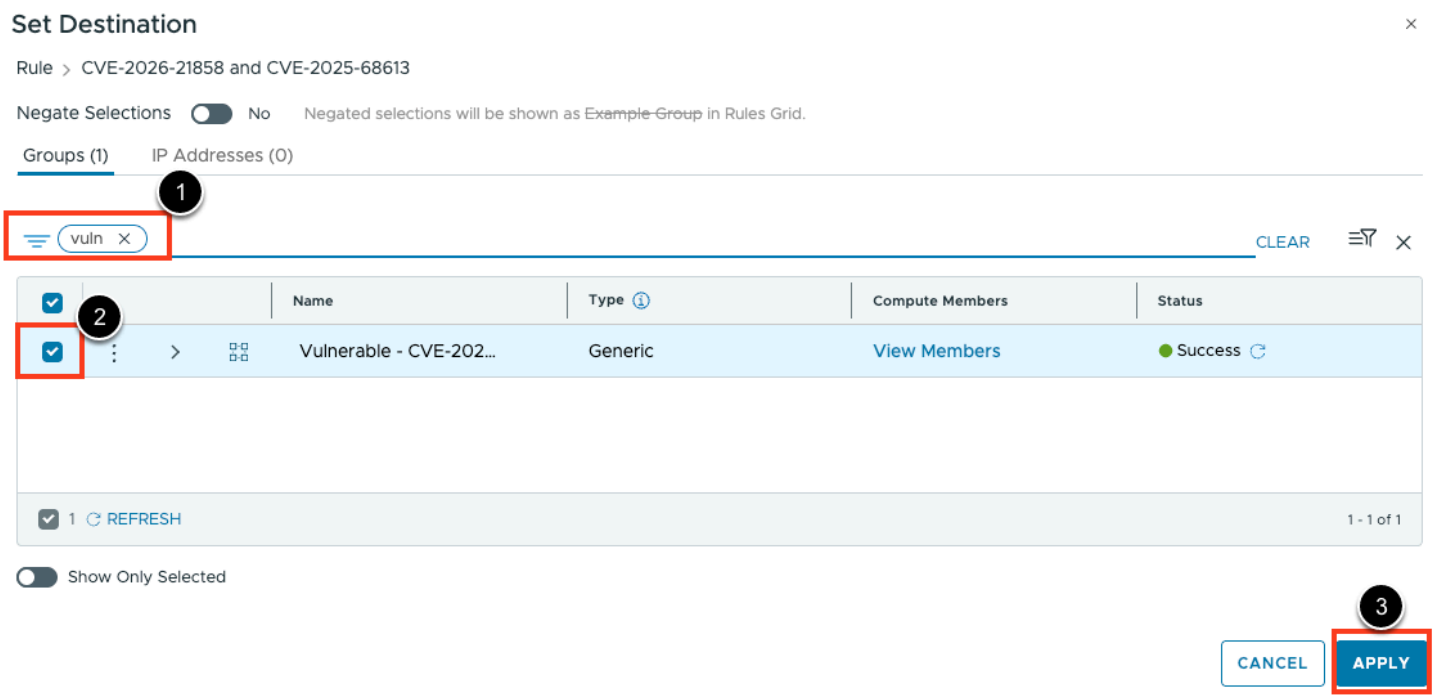
1. Click on the **3 dots** menu to the left of the **Virtual Patching** policy
2. Select **Add Rule**

Name the Rule



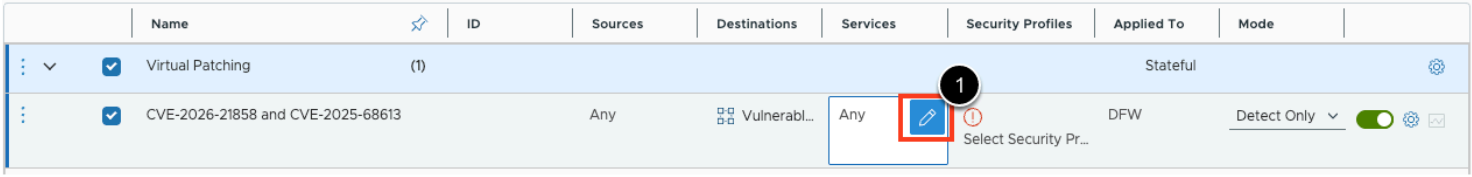
1. Click the **Name** field and type, **CVE-2026-21858 and CVE-2025-68613,** and press **enter**
2. Click in the **Destinations** field and click on the **pencil**

Set Rule Destination



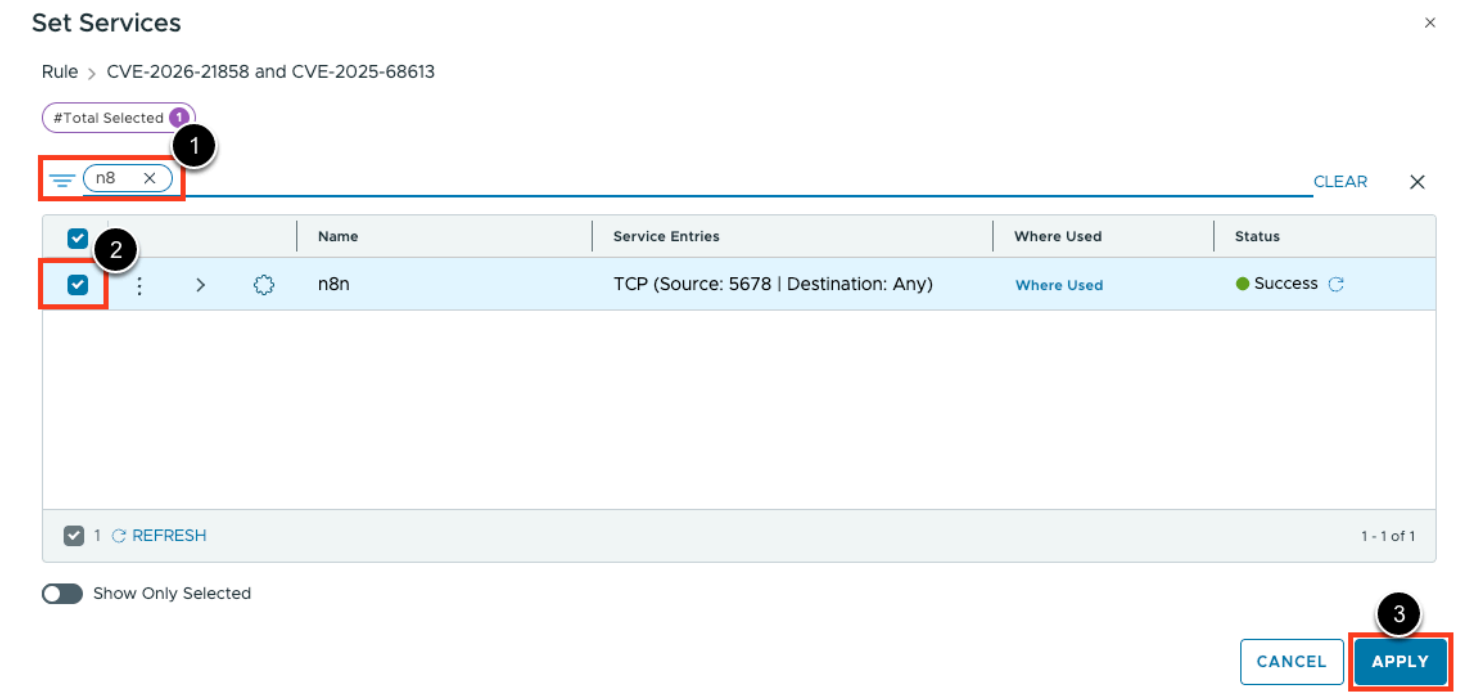
1. Click in the **Apply Filter** box, type **vuln**, and press enter
2. Select **Vulnerable - CVE-2026-21858 and CVE-2025-68613**
3. Click **APPLY**

Set Rule Services



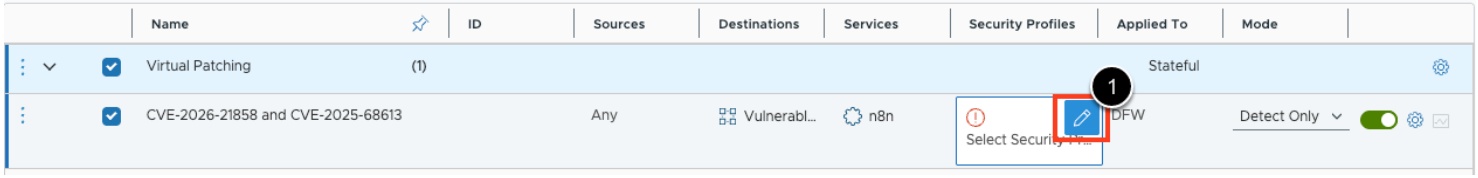
1. In the **Services** field, click on the **pencil** icon

Set Rule Services



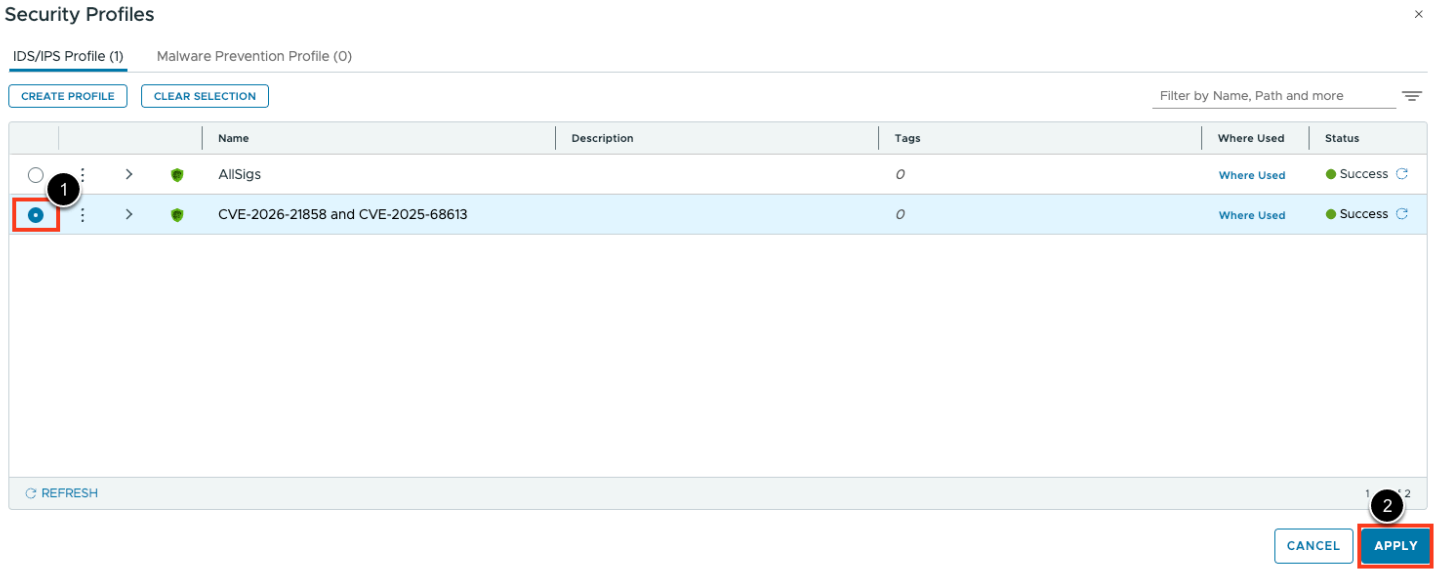
1. Click in the **Apply Filter** box, type **n8n**, and press **enter**
2. Select the **n8n** service
3. Click **APPLY**

Set Rule Security Profile



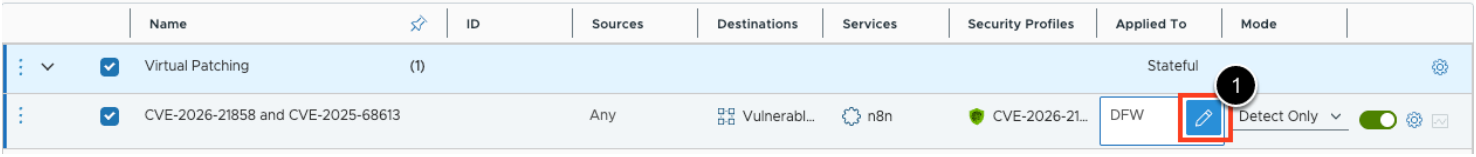
1. In the **Security Profiles** field, click on the **pencil** icon

Set Rule Security Profile



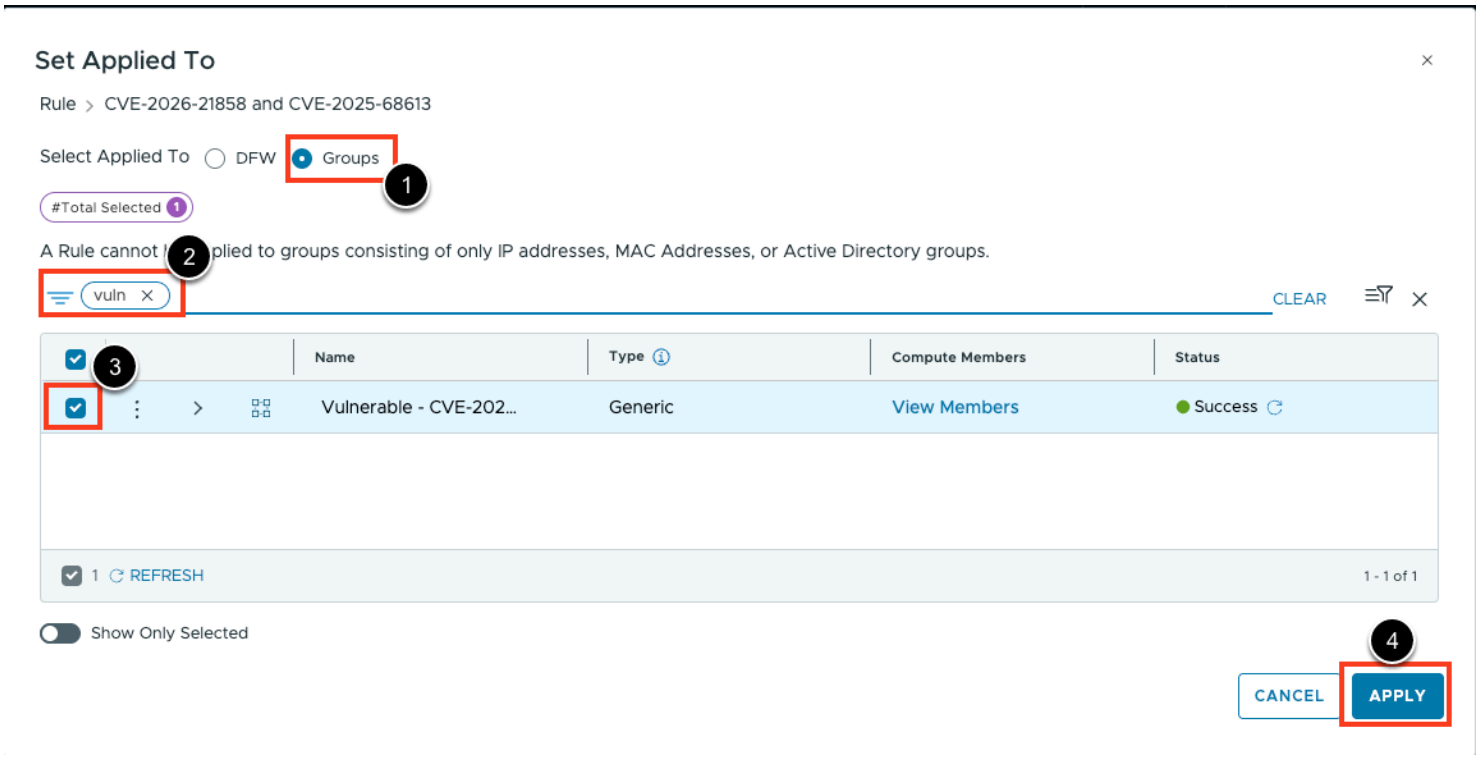
1. Select **CVE-2026-21858 and CVE-2025-68613**
2. Click **APPLY**

Set Rule Apply To



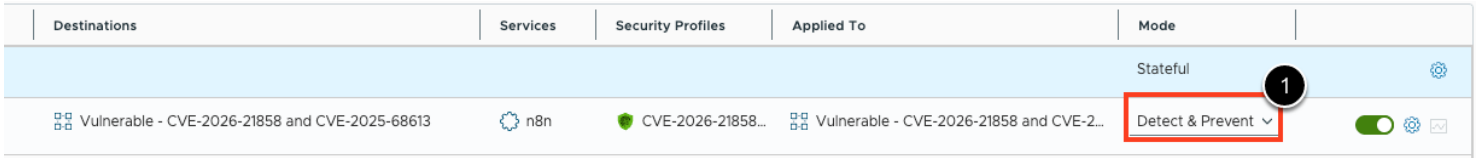
1. In the **Applied To** field, click the **pencil** icon

Set Rule Apply To



1. Click **Groups**
2. Click in the **Apply Filter** box, type **vuln**, and press **enter**
3. Select **Vulnerable - CVE-2026-21858 and CVE-2025-68613**
4. Click **APPLY**

Set Rule Mode to Detect & Prevent



1. In the **Mode** field, select **Detect & Prevent**

Review Virtual Patching Rule

Name = **CVE-2026-21858 and CVE-2025-68613**

Sources = **Any**

Destinations = **Vulnerable - CVE-2026-21858 and CVE-2025-68613**

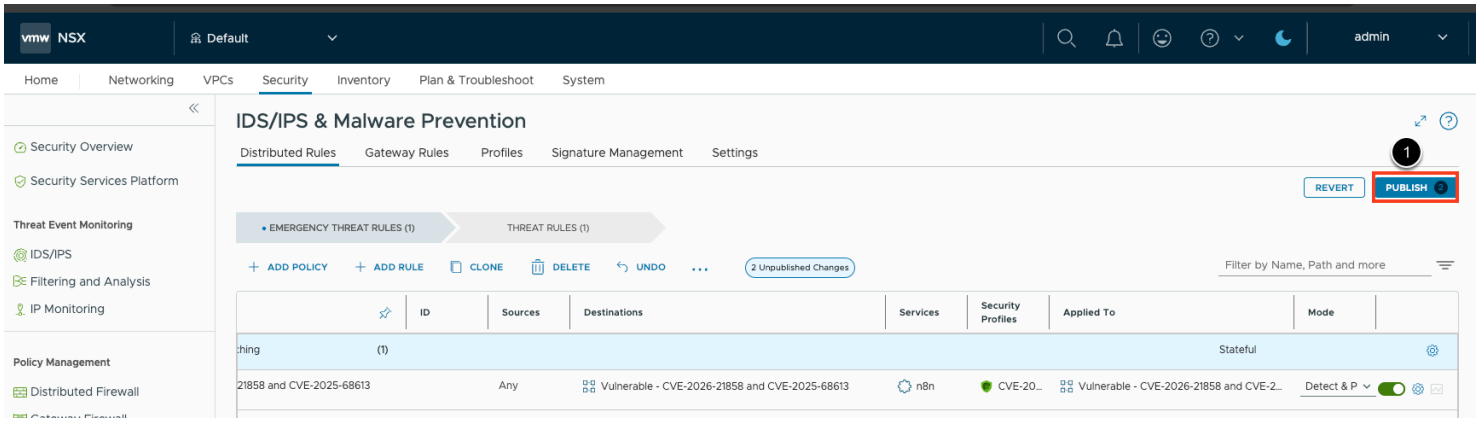
Services = **n8n**

Security Profiles = **Virtual Patch: CVE-2026-21858 and CVE-2025-68613**

Applied To = **Vulnerable - CVE-2026-21858 and CVE-2025-68613**

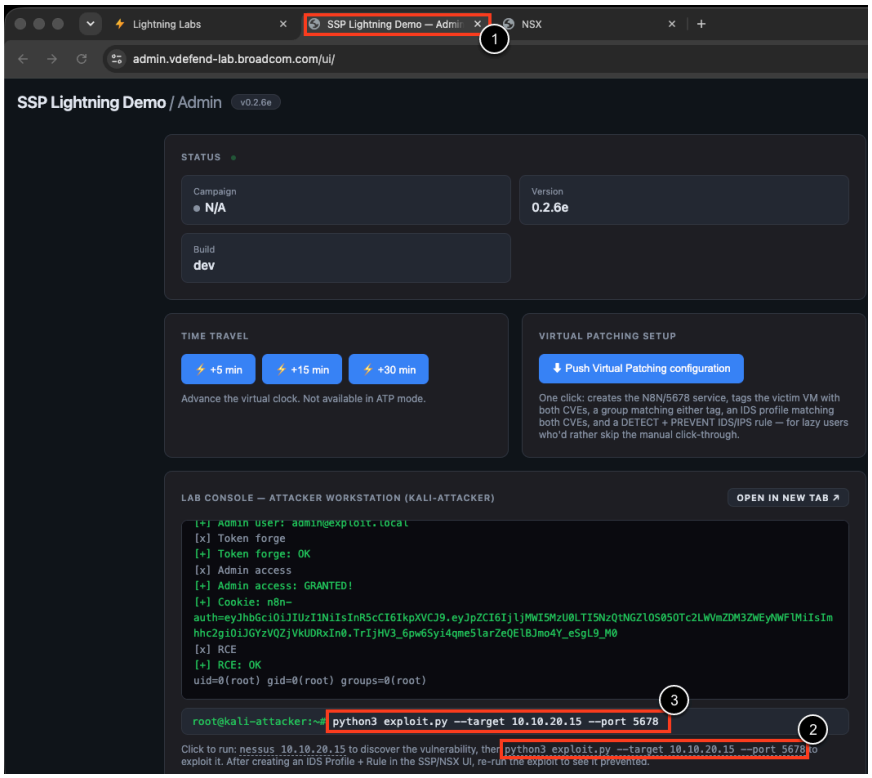
Mode = **Detect & Prevent**

Publish the Virtual Patching Policy & Rule



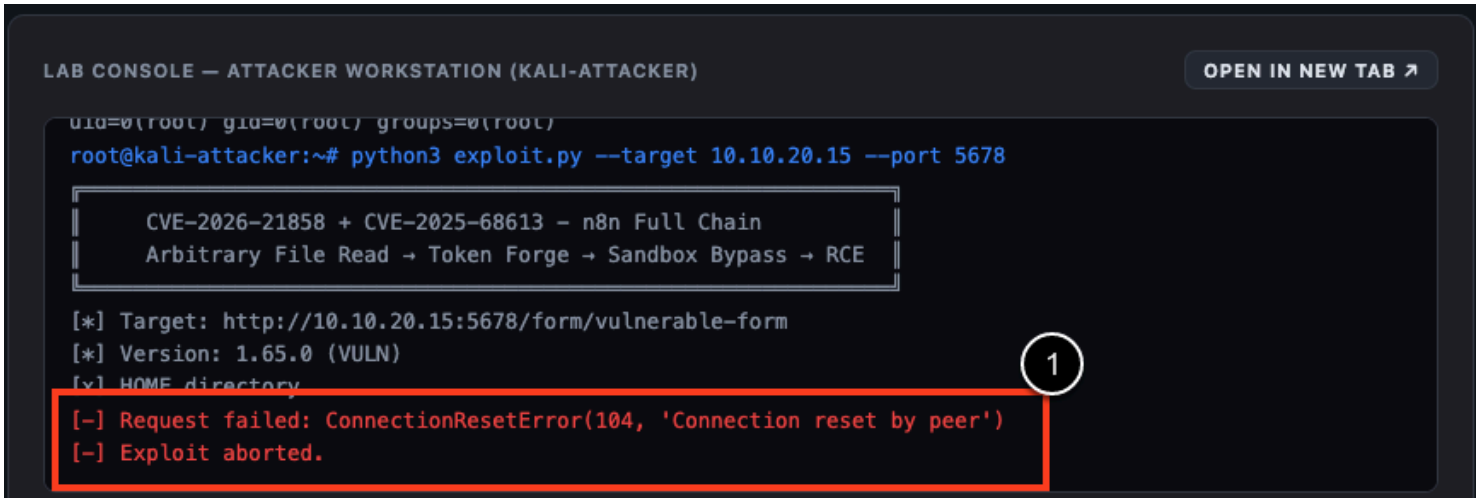
1. Click **PUBLISH**

Test the IDS/IPS Policy



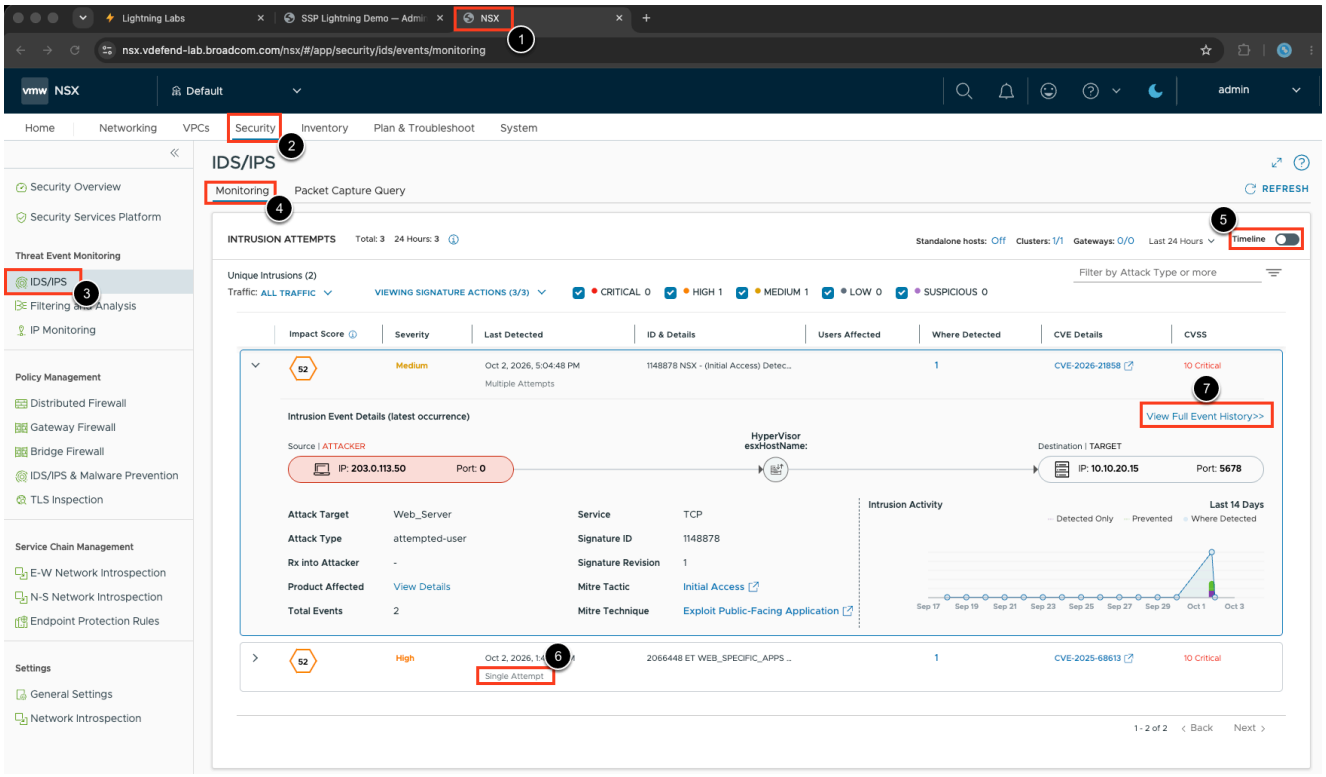
1. Switch to the **SSP Lightning Demo / Admin** tab
2. **Click to run: python3 exploit.py --target 10.10.20.15 --port 5678** (this will auto type the command into the console)
3. Review the **command** in the console and press **enter** (this will rerun the exploit)

Review Exploit Failure



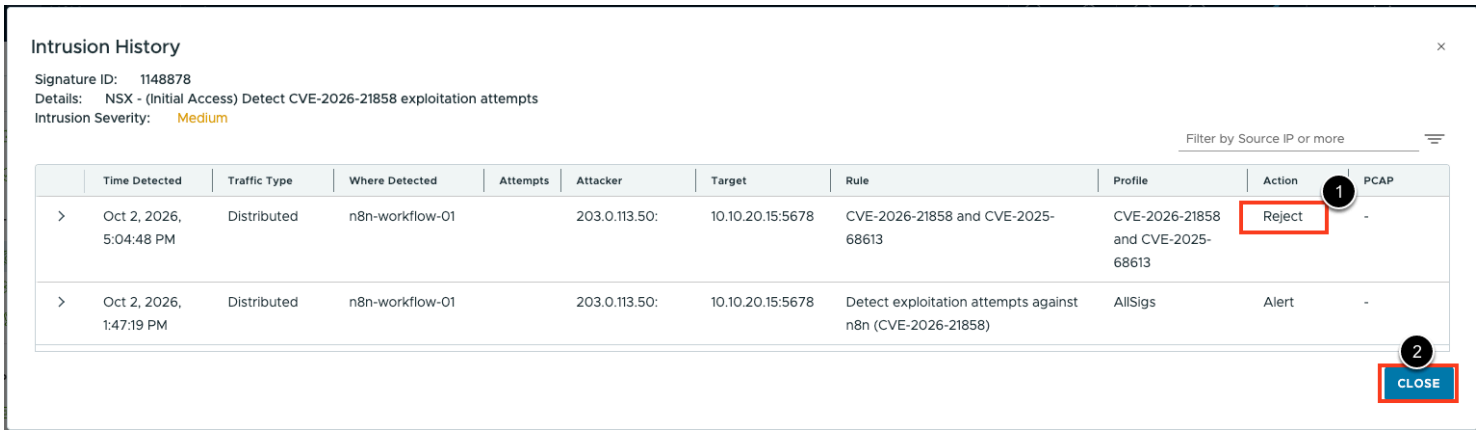
1. Note that the exploit can longer connect to our **n8n** VM

Review Exploit Attempt in NSX Manager



1. Switch to **NSX Manager**
2. Click **Security**
3. Click **IDS/IPS**
4. Click **Monitoring**
5. Turn off the **Timeline** to create some additional room on the screen
6. Note: The second signature was not triggered a second time. The attack was stopped by the first signature thus never allowing the attacker to run the second part of the attack chain!
7. Click **View Full Event History**

Intrusion History



1. Note that the latest exploit attempt was **rejected** by our policy
2. Click **CLOSE**

This concludes the Virtual Patching lab!